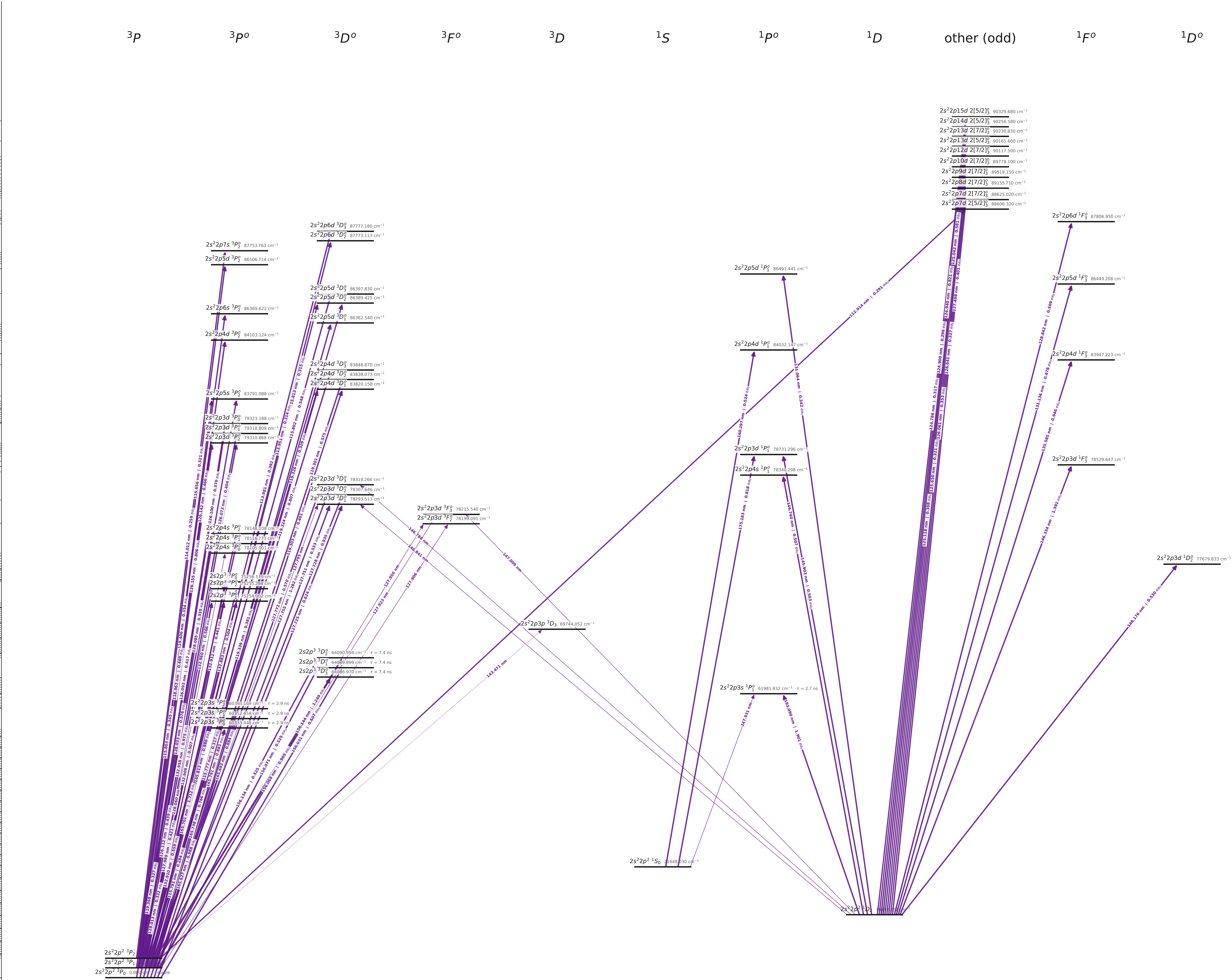


C (C I). Strongest classified transitions below 2 μm (NIST): 59 levels, 80 lines.

C^+ ionisation limit $90820.35 \text{ cm}^{-1} = 11.26029 \text{ eV}$ (110.107 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle 0 || e r || j \rangle$ in $e a_0$
 Matrix elements are measured or NIST values; * = high-accuracy theory, = model calculation.
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e J' \rangle$ (ea)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$2s^22p^2\ ^3P_2 - 2s^22p3p\ ^3D_3$	143.4707	-	2089.5726	-	-	-	-	two-photon excitation of ground-state carbon atoms at 2 x ...
$2s^22p^2\ ^3P_1 - 2s^22p3s\ ^3P_2^o$	165.6267	-	1810.0495	0.988	8.700e+07	54.88	25.2	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
$2s^22p^2\ ^3P_0 - 2s^22p3s\ ^3P_1^o$	165.6928	-	1809.3272	0.884	1.160e+08	54.88	33.6	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
$2s^22p^2\ ^3P_2 - 2s^22p3s\ ^3P_2^o$	165.7008	-	1809.2402	1.712	6.10e+08	54.88	75.7	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
$2s^22p^2\ ^3P_1 - 2s^22p3s\ ^3P_0^o$	165.7379	-	1808.8350	0.766	8.700e+07	54.88	25.2	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
$2s^22p^2\ ^3P_1 - 2s^22p3s\ ^3P_2^o$	165.7907	-	1808.2591	0.883	3.470e+08	54.88	100	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
$2s^22p^2\ ^3P_2 - 2s^22p3s\ ^3P_1^o$	165.8121	-	1808.0257	0.986	1.440e+08	54.88	41.8	165.7 nm resonance multiplet 2p2 3P - 2p3s 3P: measured f...
$2s^22p^2\ ^3D_2 - 2s^22p3s\ ^3P_2^o$	193.0905	-	1552.6004	1.901	3.390e+08	58.95	91.5	193.09 nm line: carbon line used in white-dwarf and comet...
$2s^22p^2\ ^1D_2 - 2s^22p3s\ ^1P_1^o$	247.9310	247.8561	1209.1770	-	-	58.95	-	2478.56 Å

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

^{13}C hyperfine structure ($I = 1/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$2s^2 2p^2 \ ^3P_1$	2.838(17)	-	F=1/2: -2.84 F=3/2: +1.42
$2s^2 2p^2 \ ^3P_2$	149.055(10)	-	F=3/2: -223.58 F=5/2: +149.06

Measured A, B constants; sources in the data file

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$2s^2 2p^3 \ ^3P_1 - 2s^2 2p^3 s \ ^3P^o_2$	165.627	13-12	-3028(1049)
$2s^2 2p^3 \ ^3P_0 - 2s^2 2p^3 s \ ^3P^o_1$	165.693	13-12	-3418(1049)
$2s^2 2p^3 \ ^3P_2 - 2s^2 2p^3 s \ ^3P^o_2$	165.701	13-12	-2908(1049)
$2s^2 2p^3 \ ^3P_1 - 2s^2 2p^3 s \ ^3P^o_1$	165.738	13-12	-2578(1049)
$2s^2 2p^3 \ ^3P_2 - 2s^2 2p^3 s \ ^3P^o_1$	165.791	13-12	-3058(1049)
$2s^2 2p^3 \ ^3P_2 - 2s^2 2p^3 s \ ^3P^o_1$	165.812	13-12	-3118(1049)
$2s^2 2p^3 \ ^1S_0 - 2s^2 2p^3 s \ ^1P^o_1$	247.931	13-12	-4677(60)
$2s^2 2p^3 \ ^1S_0 - 2s^2 2p^3 s \ ^1P^o_1$	247.931	14-12	-8814(60)

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope})$$